1. You can create backups using “rsync” command with “-av” option.

```
kmin@kali-rrc:~$ mkdir backups
kmin@kali-rrc:~$ rsync -av importantstuff backups
sending incremental file list
importantstuff/
importantstuff/case.sh
importantstuff/dataGen.sh
importantstuff/elseif.sh
importantstuff/file0example.txt
importantstuff/file1example.txt
importantstuff/file2example.txt
importantstuff/file3example.txt
importantstuff/file4example.txt
importantstuff/file5example.txt
importantstuff/file6example.txt
importantstuff/file7example.txt
importantstuff/file8example.txt
importantstuff/file9example.txt
importantstuff/interfaces.sh
importantstuff/ipchanger.sh
importantstuff/logfile.txt
importantstuff/my_first_script.sh
importantstuff/output.txt
importantstuff/robust_ipchanger.sh
importantstuff/rps.sh
importantstuff/sedoutput.txt

sent 9,552 bytes  received 419 bytes  19,942.00 bytes/sec
total size is 7,948  speedup is 0.80
kmin@kali-rrc:~$
```

2. This “rsync” command only copies the new file, unlike the “cp” command.

```
kmin@kali-rrc:~$ mkdir importantstuff/logfile2.txt
kmin@kali-rrc:~$ rsync -av importantstuff backups
sending incremental file list
importantstuff/
importantstuff/logfile2.txt/

sent 748 bytes  received 24 bytes  1,544.00 bytes/sec
total size is 7,948  speedup is 10.30
kmin@kali-rrc:~$
```

3. But, it won’t delete the existing backup file if the user deletes the original file.

```
kmin@kali-rrc:~$ rsync -av importantstuff backups
sending incremental file list
importantstuff/

sent 705 bytes  received 20 bytes  1,450.00 bytes/sec
total size is 7,948  speedup is 10.96
kmin@kali-rrc:~$ ls backups/
importantstuff
kmin@kali-rrc:~$ ls backups/importantstuff/
case.sh     file0example.txt  file3example.txt  file6example.txt  file9example.txt  logfile2.txt        output.txt          sedoutput.txt
dataGen.sh  file1example.txt  file4example.txt  file7example.txt  interfaces.sh     logfile.txt         robust_ipchanger.sh
elseif.sh   file2example.txt  file5example.txt  file8example.txt  ipchanger.sh      my_first_script.sh  rps.sh
kmin@kali-rrc:~$
```

4. If you want to keep the current files in the backup folder without having deleted files, you can use "—delete".

```
kmin@kali-rrc:~$ rsync -av --delete importantstuff backups
sending incremental file list
deleting importantstuff/logfile2.txt/

sent 702 bytes  received 49 bytes  1,502.00 bytes/sec
total size is 7,948  speedup is 10.58
kmin@kali-rrc:~$
```

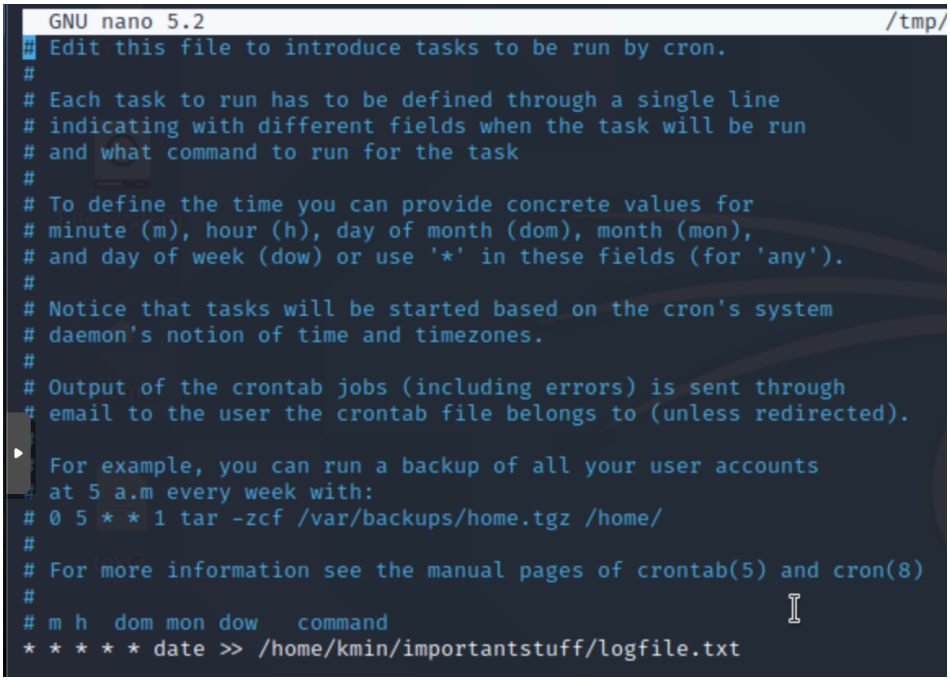
5. You can use ssh to do a backup through the network. You may need to put the option "-e" to execute.

```
kmin@kali-rrc:~$ rsync -av -e ssh importantstuff root@192.168.1.237:/root/backups
root@192.168.1.237's password:
sending incremental file list
importantstuff/
importantstuff/case.sh
importantstuff/dataGen.sh
importantstuff/elseif.sh
importantstuff/file0example.txt
importantstuff/file1example.txt
importantstuff/file2example.txt
importantstuff/file3example.txt
importantstuff/file4example.txt
importantstuff/file5example.txt
importantstuff/file6example.txt
importantstuff/file7example.txt
importantstuff/file8example.txt
importantstuff/file9example.txt
importantstuff/interfaces.sh
importantstuff/ipchanger.sh
importantstuff/logfile.txt
importantstuff/my_first_script.sh
importantstuff/output.txt
importantstuff/robust_ipchanger.sh
importantstuff/rps.sh
importantstuff/sedoutput.txt

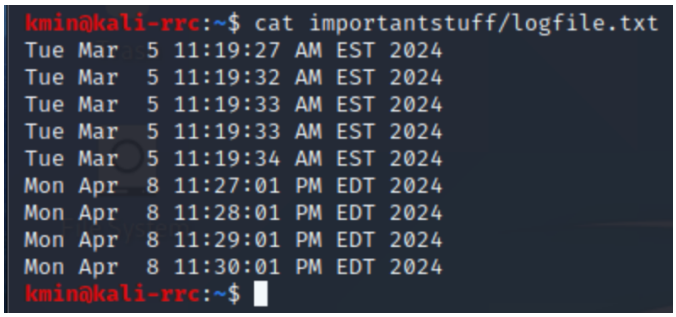
sent 9,554 bytes  received 419 bytes  2,849.43 bytes/sec
total size is 7,948  speedup is 0.80
kmin@kali-rrc:~$
```

```
[root@localhost ~]# ls backups/importantstuff/
case.sh     file0example.txt  file3example.txt  file6example.txt  file9example.txt  logfile.txt         robust_ipchanger.sh
dataGen.sh  file1example.txt  file4example.txt  file7example.txt  interfaces.sh     my_first_script.sh  rps.sh
elseif.sh   file2example.txt  file5example.txt  file8example.txt  ipchanger.sh      output.txt          sedoutput.txt
[root@localhost ~]#
```

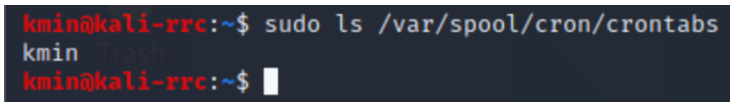
6. You can use the “crontab -e” command to automatically back up your data.



7. You may need to restart your service using the “systemctl” command. Now, your logfile may run date command automatically every minute.



8. A sudo user can see who run the crontab by viewing a special file path as shown.



9. Now, let's run a script inside cron. We may only use the method shown above to perform this.

